

A Base learners and information

Input 33 patient/context variables + 18 historical non-ARI facility features
LightGBM learning rate 0.05; 127 leaves; minimum leaf size 80; L2 = 5
XGBoost learning rate 0.05; depth 8; minimum child weight 5; L2 = 5
Both: row/feature fraction 0.8; at most 4,000 rounds; 10% early-stop holdout; patience 100

B Second layer fitted on nested base predictions

2a Stratified logistic stacking clip probabilities to $[10^{-7}, 1 - 10^{-7}]$; $C = 1000$
 $z_i^{(m)} = \text{logit}(p_i^{(m)})$, $z_i^A = \alpha + bc_i + dv_i + \sum_m (a_m + b_m c_i + d_m v_i) z_i^{(m)}$
Child and COVID-19 interactions allow population-specific intercepts and member weights.

2b Shallow residual correction $z_i^B = z_i^A + h(u_i, z_i^A)$
8 leaves; minimum leaf size 1,000; learning rate 0.03; L2 = 20; row/feature fraction 0.8
At most 500 rounds; 15% of second-layer records for early stopping; patience 30
 u_i includes demographic, infection, facility, geographic and calendar variables.

2c Unpenalised marginal logistic projection
 $p_i^H = \sigma(\beta_0 + \beta_1 z_i^B + \sum_k \gamma_k g_k(i))$, $\sum_i g_k(i)(y_i - p_i^H) \approx 0$ on fitting data
Age, sex, ethnicity, urban/rural residence, sector and infection type; year added in distribution
All second-layer records used; fitting constraints require a finite, converged solution.

C Temporal maintenance only

Four-dimensional monthly state $z_i^H = \text{logit}(p_i^H)$; $\phi_i = [1, z_i^H, v_i, c_i]^\top$
 $\text{logit}(p_{i,t}) = \phi_i^\top \theta_t$, $\theta_t = \theta_{t-1} + \varepsilon_t$, $\varepsilon_t \sim N(0, qI)$
Intercept, score slope, COVID-19 offset and child offset; $q = 0.1$; initial covariance $0.05I$
Previous state → predict month t → outcomes become available → Laplace update
Chosen by 2018–2021 forward log loss; at most 25 Newton steps, tolerance 10^{-8} .

D Three evaluation protocols with distinct information conditions

In-distribution 5 stratified outer folds For each test fold: train 3 of the other 4; predict the fourth 20 inner–outer combinations per base learner Fit second layer; test outer fold	Temporal Develop in 2015–2021 Test in 2022–2023 Internal 5-fold second-layer fitting Compare static, rolling and matched state updates Reporting delay not simulated	Geographic 6 province-grouped folds Remove location identifiers No profiles: LGB/XGB/TabM Profiles: LGB/XGB only No full second layer Local historical data available
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E Comparators and outcome measures

Comparators LR-7; equal-weight LightGBM/XGBoost/TabM ensemble (EW)
Discrimination AUROC; AUPRC (average precision) **Probability error** log loss; Brier score
Calibration ICI; weighted mean/max $|\log(O/E)|$ across 42 marginal groups; worst eligible-group ECE
Review prioritisation decision curves and mortality coverage; paired record-level bootstrap

c_i indicates children; v_i indicates COVID-19; m indexes base learners. Historical aggregates are precomputed. Projection shares outcomes with residual early stopping. Direct fitting separation does not isolate every historical label, and fitting constraints do not guarantee test-group calibration. Geographic comparisons change both profiles and learner membership.